# RansomWatch: Early Ransomware Detection System

## Comprehensive Architecture, Multi-Stage Pipeline & Operational Workflow Guide

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| **Project:** Early Ransomware ML + Explainable Dashboard | **Foundation:** MLRan (JNCA 2026) + RanSMAP | **Lead Contributor:** Kalyani (Member 3) | **Status:** Production Ready (100% Track A) |

## 1. Problem Statement & Design Objectives

Modern ransomware (e.g., LockBit, Conti, BlackCat) completes user-file encryption in under **30 to 60 seconds**. Traditional signature-based antivirus and monolithic machine-learning classifiers suffer from two fatal operational flaws:
1. **Late Detection Latency:** Standard post-execution analyzers wait for execution completion (~120s), resulting in 100% data loss prior to alerting.
2. **False Positive Disruption:** Legitimate cryptographic applications (e.g., AESCrypt, VeraCrypt, 7-Zip) write high-entropy scrambled data, causing naive entropy filters to raise destructive false alarms.

**RansomWatch solves this through a Two-Stage Detection Architecture (D-005):** rapidly triggering soft process suspension within **18.0 seconds** via sub-second storage block telemetry, followed by high-precision behavioral confirmation that eliminates false alarms on benign cryptographic utilities (**0% false alarm rate on AESCrypt**).

## 2. The End-to-End Operational Workflow

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| **Stage 1: Multi-Modal Sensor Ingestion** | • **Storage Block Telemetry (RanSMAP):** Continuous ATA sector writes, Logical Block Addresses (LBA locality), write size, and Shannon block entropy.<br>• **API Sequence Interception (Cuckoo):** Chronological API stream capturing FindFile → Open → Write → Delete/Rename lifecycle sequences.<br>• **Canary / Honeyfile Tripwires:** Pre-deployed bait files in Desktop/Documents; triggers immediate high-confidence alarms upon modification.<br>• **Flow-Level Network Signals:** DNS queries, DGA entropy, connection burst rate, and dead host C2 connection failures. |
| **Stage 2: Real-Time Streaming Alert (Fast Containment)** | • **Sliding Window Engine:** Segmented 2.0-second time windows with 1.0-second stride extracting 31 sequential trajectory features.<br>• **EWMA Exponential Smoothing:** Smoothing parameter $\alpha$=0.3, decision threshold $\theta$=0.5; filters isolated transient disk spikes.<br>• **Ultra-Fast Time-to-Detection (TTD):** Triggers in **18.0 seconds median** (100% ransomware catch rate).<br>• **Soft Containment:** Suspends suspicious thread execution and freezes volume shadow copies before mass file destruction occurs. |
| **Stage 3: Holistic Confirmation & False-Positive Elimination** | • **Two-Stage Architecture (D-005):** The holistic Stage 2 classifier evaluates static PE headers, imported DLLs, mutexes, and registry persistence.<br>• **Benign Crypto Safe Dismissal:** Legitimate encryption utilities (AESCrypt, VeraCrypt) flagged by Stage 1 are verified as benign and resumed (**0% FPR**).<br>• **Hard Containment:** Confirmed ransomware processes are terminated, binary artifacts quarantined, and an incident ticket auto-dispatched. |
| **Stage 4: Explainable AI (SHAP) & SOC Triage** | • **Sub-20ms SHAP Service:** Computes linear Shapley values in <0.02 seconds, translating raw numeric vectors into plain-language reasons.<br>• **MITRE ATT&CK; Tagging:** Maps firing indicators to T1486 (Data Encrypted for Impact), T1055 (Process Injection), and T1490 (Inhibit System Recovery).<br>• **RansomWatch Dashboard:** Web console featuring Live Sample Inspector, DVR-style Execution Replay, and Alert Management. |
| **Stage 5: Active Learning & Edge ONNX Deployment** | • **Human-in-the-Loop Feedback:** Analysts review borderline alerts (uncertainty sampling near p=0.5); corrections retrain candidate models.<br>• **Concept Drift Hardening:** Model retraining with sample weighting counteracts modern goodware drift (AUT = 98.54%).<br>• **ONNX Runtime Export:** Highly optimized engine (<100 KB model footprint, <1.0 ms CPU inference) for native endpoint deployment. |

## 3. Quantitative Performance & Ablation Highlights

Extensive empirical evaluation across 16 experiments (traced in memory/09_EXPERIMENT_LOG.md) demonstrates that RansomWatch dramatically outperforms the baseline MLRan paper and standalone window classifiers across all operational metrics:

| Architecture / Model | Balanced Acc | Recall (Catch) | Benign FPR | Median TTD | AESCrypt FPR |
|---|---|---|---|---|---|
| **MLRan Paper Baseline (LR)** | 98.18% | 98.71% | 2.35% | 120.0s (End) | N/A |
| **Stage 1 Only (Streaming EWMA)** | 85.28% | 100.0% | 11.1% | 18.0s | 100.0% (Fail) |
| **Stage 2 Only (Full Post-Execution)** | 98.18% | 100.0% | 0.0% | 120.0s (Late) | 0.0% |
| **RansomWatch Two-Stage (Ours)** | **98.85%** | **100.0%** | **0.00%** | **18.0s** | **0.0% (Clean)** |

## 4. RansomWatch Interactive Dashboard Console

The SOC dashboard (dashboard/app.py) provides an interactive triage interface for security responders:

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| **Tab 1: Live Sample Inspector** | Upload suspicious PE binaries or select test samples; renders animated verdict banner (RANSOMWARE vs BENIGN), calibrated probability gauge, and top-5 plain-language SHAP attribution cards with MITRE ATT&CK; technique tags. |
| **Tab 2: Execution Replay (DVR)** | Interactive timeline scrubber ($T=0$ to $120$s) replaying live disk I/O, sector write bursts, entropy progression curves, exact alert trigger markers ($TTD=18$s), and calculated user-data megabytes saved from encryption. |
| **Tab 3: Threat Alert Database** | Persistent SQLite threat table storing detection records, SHA-256 hashes, timestamps, and triage workflows (QUARANTINED, INVESTIGATING, DISMISSED) with one-click CSV export and active-learning feedback triggers. |
| **Tab 4: Model Card & Trust** | Complete transparency reporting: 8-model chronological benchmark, concept drift tracking across 2020–2024 submission years, ONNX endpoint resource measurements, and acknowledged threats to validity. |

## 5. Verification & Reproducibility

• **Full Test Suite:** 69 / 69 passing green via pytest covering data loaders, streaming rules, adversarial attacks, ONNX export, and explanation engines.
• **Launch Dashboard:** .\.venv\Scripts\streamlit run dashboard/app.py (Active at http://localhost:8501).
• **Full Traceability:** Every experiment metric traces to memory/09_EXPERIMENT_LOG.md and reports/final_tables.md.